日程表 Agenda

November 6, 2026（Friday） 南京东方珍珠 voco 酒店　大堂 Nanjing voco Oriental Pearl Hotel, Lobby	
下午 Afternoon	**注册报到 Registration**
17:30	**晚　餐 (沙塘园食堂)** **Dinner (Shatangyuan Canteen)**

November 7, 2026（Saturday） 东南大学四牌楼校区　吴健雄纪念馆报告厅 Chien-Shiung Wu Memorial Hall, Southeast University			
主持人 Chair： 江云峰 JIANG Yunfeng			
8:55—9:00	**开场欢迎 Opening（杨文力 YANG Wen-Li）**		
9:00—9:45	管习文 GUAN Xi-Wen	Fractional Exclusion Statistics in Quantum Holonomy	中科院精密测量科学与技术创新研究院 APM, Chinese Academy of Sciences
9:45—10:30	吴建达 WU Jianda	Many-body quantum geometry in time-dependent systems with instantaneous quantum integrable field theory	同济大学 Tongji University
10:30—10:45	**茶歇、合影 Tea Break and Photo taking**		
10:45—11:30	赵宏政 ZHAO Hongzheng	Controlling heating: from Floquet prethermalization to Floquet superheating	北京大学 Peking Universtiy
11:30—12:15	程　颂 CHENG Song	Spectral Signatures of Anyonic Statistics in One Dimension	香港大学 Hongkong University
12:15—14:00	**午 餐 (沙塘园食堂)　Lunch (Shatangyuan Canteen)** **午 休 Noon Break**		

主持人 Chair：吴俊宝 WU Jun-bao			
14:00—14:45	束红非 SHU Hongfei	From fractional WKB to graded TBA System	郑州大学 Zhengzhou University
14:45—15:30	何　淼 HE Miao	A universal scaling function for giant graviton OPE coefficients	安徽师范大学 Anhui Normal University
15:30—15:45	**茶 歇 Tea Break**		
15:45—16:30	谷　伟 GU Wei	Virasoro Constraints from BF Reduction in JT Gravity	浙江大学 Zhejiang University
16:30—17:15	PRYBITOK Anton	Non-Semisimple Integrability and Dynamical Limits	北京雁栖湖应用数学研究院 BIMSA
18:00	**晚 宴 (南京东方珍珠 voco 酒店，火烈鸟厅)** **Dinner Banquet (Flamingo, Nanjing voco Oriental Pearl Hotel)**		

November 8, 2026（Sunday） **东南大学四牌楼校区　吴健雄纪念馆报告厅** **Chien-Shiung Wu Memorial Hall, Southeast University**			
主持人 Chair：杨文力 YANG Wenli			
9:00—9:45	张世忠 ZHANG Shizhong	Viscosity Drude weight of one-dimensional Bose and Fermi gases	香港大学 Hongkong University
09:45—10:30	姜玉铸 JIANG Yuzhu	Thermodynamics and Dynamics in Integrable Models of 1D Quantum Gases	中科院精密测量科学与技术创新研究院 APM, Chinese Academy of Sciences
10:30—10:45	**茶 歇 Tea Break**		
10:45—11:30	张甲举 ZHANG Jiaju	Single-eigenstate test of eigenstate thermalization via perturbed eigenstate quench	天津大学 Tianjin University
11:30—12:15	陈洋洋 CHEN Yangyang	Dynamical Correlation of the Post-quench Non-thermal Equilibrium State	西北大学 Northwest University
12:15—14:00	**午 餐 (沙塘园食堂)　Lunch (Shatangyuan Canteen)** **午 休 Noon Break**		

主持人 Chair：　吴建达 WU Jianda			
14:00—14:45	左　芬 ZUO Fen	在量子计算机上生成自旋态	上海微观纪元科技有限公司 Shanghai MiQro Era Digital Technology Co. Ltd
14:45—15:30	白　楠 BAI Nan	Integrability of flavored ABJM spin chain from ABA	广西师范大学 Guangxi Normal University
15:30—15:45	**茶 歇 Tea Break**		
15:45—16:30	乔　艺 QIAO Yi	Explicit Boundary-Entropy Deficits in the XX Spin Chain with arbitrary boundary fields	西北大学 Northwest University
16:30—17:15	郝　雨 HAO Yu	Quantum Entanglement of Bethe states	东南大学 Southeast University
17:15—17:30	总　结 Closing		
18:00	**晚　餐 (沙塘园食堂)** **Dinner (Shatangyuan Canteen)**		

November 9, 2026（Monday）	
上午 Morning	**离会 Departure**

报告题目与摘要

Titles and Abstracts

管习文 GUAN Xi-Wen

中国科学院精密测量科学与技术创新研究院
Innovation Academy for Precision Measurement Science and Technology, CAS

题目 Title：Fractional Exclusion Statistics in Quantum Holonomy

摘要 Abstract：

There are fundamental differences between the properties of bosons and fermions, which obey Bose–Einstein (BE) and Fermi–Dirac (FD) statistics, respectively. A common question is whether these constitute the only possible forms of quantum statistics found in nature. We now know that in two dimensions, particles can exhibit anyonic statistics, which interpolate continuously between BE and FD statistics. Going further, Haldane formulated fractional exclusion statistics (FES), a description that enables continuous interpolation between the two statistics in arbitrary spatial dimensions. Nevertheless, theoretical characterization and experimental observation of such fractionalized states remain highly challenging.

In this talk, we demonstrate that topological energy pumping in one-dimensional interacting two-component Fermi and Bose gases offers a controlable route to realize fractionalized states obeying FES. In particular, full transmutability between dynamical and statistical interactions leads to an emergent segmentation of states with distinct FES near the zero-interaction transition in both models. We show that at this zero-interaction crossing, a subtle shift in the quantum numbers of both spin and charge degrees of freedom occurs abruptly, while the total momentum remains unchanged. This behavior directly reflects the nonanalytic nature of the energy at the zero-interaction point and reveals how internal degrees of freedom reshape the segmented macroscopic statistical distribution. Such segmented FES splitting produces a beating pattern in the one-body correlation function, in contrast to the Luttinger-liquid result at infinitely strong interaction. As a consequence, this beating pattern serves as a precise probe of the local magnetic field within topologically excited states.

吴建达 WU Jianda

同济大学 Tongji University

题 目 Title ： Many-body quantum geometry in time-dependent systems with instantaneous quantum integrable field theory

摘要 Abstract：

We study quantum geometric effects in instantaneously quantum-integrable time-dependent systems. We establish a theorem stating that the Berry connection matrix thus all

associated geometric quantities of the system can be precisely characterized by excitations up to two particles from the initial quantum integrable system. To illustrate the many-body geometric influence, we analyze an Ising chain subjected to both a small longitudinal field and a slowly rotating transverse field, whose low-energy physics in the scaling limit is instantaneously governed by the quantum E8 integrable field. Focusing on the quantum geometric potential (QGP), we show the QGP continuously suppresses the instantaneous energy gaps with decreasing longitudinal field, thereby enhancing many-body Landau-Zener tunneling as evidenced by the Loschmidt echo and its associated spectral entropy. The critical threshold for the longitudinal field strength is determined, where the spectral entropy linearly increases with system size and exhibits hyperscaling behavior when approaching to the threshold. As the longitudinal field passes the threshold and decreases toward zero, the QGP continuously leads to vanishing instantaneous energy gaps involving more low-energy excitations, resulting in increasing spectral entropy indicative of many-body Landau-Zener tunneling. Our results unveil telltale quantum geometric signatures in time-dependent many-body systems, elucidating the intricate interplay between quantum geometry and dynamics in quantum many-body systems.

赵宏政 ZHAO Hongzheng

北京大学 Peking University

题目 Title：Controlling heating: from Floquet prethermalization to Floquet superheating

摘要 Abstract：

Present-day quantum computers provide an ideal platform for studying periodically driven quantum systems. Such systems, however, generically absorb energy and heat towards a featureless infinite-temperature state. This talk addresses a central question: how can Floquet heating be controlled? We discuss several mechanisms for suppressing heating, including spectral engineering and its experimental realization on a superconducting quantum processor. Finally, we introduce Floquet superheating, a dynamical analogue of the superheating phenomenon familiar from first-order phase transitions.

程颂 CHENG Song

香港大学 Hongkong University

题目 Title：Spectral Signatures of Anyonic Statistics in One Dimension

摘要 Abstract：

Exchange statistics is a basic ingredient of quantum many-body physics, yet its dynamical consequences can be difficult to separate from those of interactions. In this talk, I will address this problem by discussing the spectral function of the repulsive anyonic Lieb–Liniger gas. Combining Bethe-ansatz results with mobile-impurity theory, we determine the power-law singularities on the spectral threshold. By comparing corresponding dressed excitations at

fixed effective coupling, the effect of exchange statistics on the spectral response is isolated, and we predict that statistics can turn a nonsingular particle-removal contribution into a divergent one without creating a new excitation branch, and thus allows a concurrence of three divergence at a single momentum cut which is forbidden in a bosonic system.

束红非 SHU Hongfei

郑州大学 Zhengzhou University

题目 Title：From fractional WKB to graded TBA System

摘要 Abstract：

The TBA/WKB correspondence is a remarkable manifestation of the ODE/IM correspondence, highlighting deep connections between quantum spectral problem, resurgence, and quantum integrable model Remarkably, the exact WKB periods of Schr\"odinger equations are governed by thermodynamic Bethe ansatz (TBA) equations of the quantum integrable models. This raises a natural question: can novel structures in exact WKB theory lead to new structures in integrable systems? In this talk, I will explore this question by showing how fractional WKB expansions for the cubic oscillator give rise to $\mathbb{Z}_5$- and $\mathbb{Z}_7$-graded Y-systems and their associated TBA equations. I will also discuss their wall-crossing behavior and the resulting connections to the graded TBA systems of the Lee–Yang model.

何淼 HE Miao

安徽师范大学 Anhui Normal University

题目 Title：A universal scaling function for giant graviton OPE coefficients

摘要 Abstract：

We consider the large spin limit $S\to\infty$ of the OPE coefficient for two maximal giant gravitons and a spinning operator with finite twist in planar $\mathcal{N}=4$ super-Yang-Mills theory. Up to an overall normalization factor, this OPE coefficient exhibits a power-law scaling of the form $S^{d(g)}$. We provide strong evidence that the exponent $d(g)$ is free from finite size corrections. Consequently, the all-loop asymptotic expression for the OPE coefficient can be used to determine $d(g)$ exactly. We compute a conjectural expression for the scaling function $d(g)$ for arbitrary values of the 't Hooft coupling g. At weak coupling, our result agrees perfectly with the available field-theoretic results up to three-loop order. At strong coupling, we present the first three orders as concrete predictions. We further provide the finite-coupling result and show that it interpolates smoothly between the weak- and strong-coupling regimes.

谷伟 GU Wei

浙江大学 Zhejiang University

题目 Title：Virasoro Constraints from BF Reduction in JT Gravity

摘要 Abstract：

It is well known that two-dimensional topological gravity and JT gravity satisfy Virasoro constraints, reflecting their underlying integrable structure. In this talk, I will discuss the Virasoro constraints in deformed JT gravity.

PRYBITOK Anton

北京雁栖湖应用数学研究院
Beijing Institute of Mathematical Sciences and Applications (BIMSA)

题目 Title：Non-Semisimple Integrability and Dynamical Limits

摘要 Abstract：

We investigate Jordan structure and dynamical limits in six families of integrable spin-1/2 chains obtained from the boost automorphism method. Local operator algebras connect nilpotent evolution on finite periodic chains to central defects, unipotent R-matrices and constant braid limits. For conservative endpoint erosion type processes, we determine the characteristic and minimal polynomials and show that the Jordan block structure can change while both polynomials remain unchanged. We also identify the deletion limits that distinguish the parity obstruction to removing signs in Class 5 from the nilpotent parity dependence in Class 6 and its implications. In this context matching expansions and fragmentation recursions are derived, which yield exact amplitudes and survival probabilities for processes admitting killing. In addition, we derive obstructions to conservative Markov representations and determine when the considered conservative local generators arise from regular difference form R-matrices. The quantum algebraic structures emerging from the corresponding quadratic relations are also analysed.

张世忠 ZHANG Shizhong

香港大学 Hongkong University

题目 Title：Viscosity Drude weight of one-dimensional Bose and Fermi gases

摘要 Abstract：

I discuss bulk viscosity in one-dimensional Lieb-Liniger Bose gas and spinless fermion gas with effective range corrections. In particular, we compute exactly the viscosity Drude weight using hydrodynamic projection within the framework of generalised hydrodynamics. We first show that general argument that a proper definition of bulk viscosity within GGE would render it identically zero. On the other hand, within the standard Gibbs ensemble, we found a

nonzero Drude weight which is consistent with those computed by diagrammatic expansion, in addition to a few new results.

姜玉铸 JIANG Yuzhu

中国科学院精密测量科学与技术创新研究院
Innovation Academy for Precision Measurement Science and Technology, CAS

题目 Title：Thermodynamics and Dynamics in Integrable Models of 1D Quantum Gases

摘要 Abstract：

Ultracold atomic systems possess excellent controllability and serve as clean platforms for studying many-body physics. With the advancement of experimental techniques, research on one-dimensional (1D) quantum integrable gas models has gradually shifted its focus toward correlation functions and dynamics. Quantum integrable theory enables us to derive rigorous analytical solutions and provide quantitative theoretical interpretations for the energy spectra, non=equilibrium dynamics, quasiparticle excitations, and quantum phase transitions of 1D quantum many-body systems. This report will present our recent research progress on the thermodynamics and dynamics of 1D integrable systems, covering the correlation functions of two-component ultracold atomic systems and the dynamical behaviors of impurity systems.

张甲举 ZHANG Jiaju

天津大学 Tianjin University

题目 Title：Single-eigenstate test of eigenstate thermalization via perturbed eigenstate quench

摘要 Abstract：

I will talk about an efficient single-eigenstate diagnostic for the eigenstate thermalization hypothesis (ETH), which we propose and numerically validate using a perturbed-eigenstate quench protocol. By introducing a weak random perturbation to an energy eigenstate to break its stationarity, we characterize the time-averaged subsystem evolution speed as a function of the subsystem-to-total system size ratio. The diagnostic relies on a robust qualitative distinction: eigenstates satisfying ETH exhibit an S-shaped curve with an inflection point near half the system size, while ETH-violating eigenstates display a convex J-shaped profile. We benchmark the criterion across paradigmatic one-dimensional spin chains covering chaotic, integrable, many-body localized, and quantum many-body scar regimes, obtaining full agreement with established thermalization phenomenology. Our method circumvents the need for explicit thermal ensemble construction, providing a robust, experimentally feasible probe of eigenstate thermalization at the single-eigenstate level.



陈洋洋 CHEN Yangyang

西北大学 Northwest University

题目 Title：Dynamical Correlation of the Post-quench Non-thermal Equilibrium State

摘要 Abstract：

After a quantum quench, the integrable system is expected to relax to a non-thermal equilibrium state (NTES) whose local properties are believed to be governed by a generalized Gibbs ensemble (GGE). Combining quench action and the form factor approach, we compute the field-field correlation in the NTES produced by an interaction quench of the Lieb-Liniger model. The spectral distribution is shown to be qualitatively different from that of a thermal equilibrium state (TES): a new dispersion branch appears whose microscopic mechanism can be traced to the algebraic decaying tail for the root density distribution function, and indicates the existence of a broader family of NTES featuring similar spectral property. (arXiv: 2601.06987)

左芬 ZUO Fen

上海微观纪元科技有限公司 Shanghai MiQro Era Digital Technology Co. Ltd

题目 Title：在量子计算机上生成自旋态

摘要 Abstract：

电子的自旋二重态天然形成一个量子比特。大量电子自旋的耦合会形成非常繁复的总自旋本征态。如何在量子计算机上复现出这些自旋态呢？这一问题尽管在量子化学和多体物理中无处不在，但却一直没有得到解答。人们只针对一两种特殊的自旋态构造出了相应的量子电路，并普遍认为一般的自旋态无法高效生成。近期，通过将现有的特定电路巧妙地推广和适配到任意自旋耦合过程，我们最终彻底解决了这一难题。 作为应用，我们用生成的自旋态求解完全图和完全二分图上的海森堡模型。

白楠 BAI Nan

广西师范大学 Guangxi Normal University

题目 Title：Integrability of flavored ABJM spin chain from ABA

摘要 Abstract：

We present an algebraic Bethe ansatz framework for the two-loop flavored ABJM spin chain. Guided by the symmetry of the model, we propose an ansatz for the operator-valued reflection matrices and determine their explicit forms by imposing the soliton-non-preserving (SNP) reflection equation. We then construct a single double-row transfer matrix. Through a detailed calculation, we show that the logarithmic derivative of the transfer matrix reproduces the complete two-loop Hamiltonian, including both bulk and boundary terms, thus providing a rigorous algebraic proof of integrability. Finally, using the analytic Bethe ansatz, we derive the transfer-matrix eigenvalues and the corresponding Bethe ansatz equations.

乔艺 QIAO Yi

西北大学 Northwest University

题目 Title：Explicit Boundary-Entropy Deficits in the XX Spin Chain with arbitrary boundary fields

摘要 Abstract：

The exact thermodynamics of the XX spin-1/2 chain with arbitrary Hermitian boundary fields is constructed via fusion techniques. The rigorous result shows that (1) each end with a nonzero transverse field contributes $+\frac12T\ln2$ to the low-temperature free energy, whereas strictly diagonal and free ends contribute none; (2) there exists a characteristic temperature for each end depending on the strength of the corresponding transverse field. Just below this temperature the boundary entropy deficit appears asymptotically. As the transverse field tends to zero, the boundary crossover scale vanishes and the low-temperature expansion becomes nonuniform. The present result provides a clear benchmark not only for the boundary thermodynamics of open boundary quantum integrable systems, but also for dot defects in quantum critical systems.

郝雨 HAO Yu

东南大学 Southeast University

题目 Title：Quantum Entanglement of Bethe states

摘要 Abstract：

This talk explores the quantum entanglement of Bethe states in integrable systems. We introduce a method for computing entanglement entropy based on the algebraic Bethe ansatz and Schmidt decomposition, and apply it to several lattice models and the Lieb–Liniger model as a continuum field theory. We investigate the relationship between entanglement extrema and Bethe-root configurations, and compare our results for the Lieb–Liniger model with analytical predictions in various limits. Through extensive numerical scans, we identify counterexamples to intuitive expectations about the relationship between energy and entanglement, particularly for bound states in certain regimes. In the Lieb–Liniger model, we also observe nontrivial entanglement behavior in the strongly repulsive and strongly attractive limits, $c\to+\infty$ and $c\to-\infty$. By relaxing the Bethe equations and optimizing over off-shell states, we further explore the attainable range of entanglement and its scaling, examining how the structure of Bethe states and the eigenstate constraints affect quantum entanglement.

会议相关信息

Related Information

1. Campus location

(1) Sipailou Campus of Southeast University（东南大学四牌楼校区）

The campus has three gates: south gate, east gate and west gate. The main gate is the south gate (南门), on Sipailou (四牌楼)。Shing-Tung Yau Center is near the east gate (东门), on Chengxian Jie (成贤街)。The west gate (西门) is on Jinxianghe Road (进香河路)。(See the blue circular marks in the map picture on the following page.)

(2) Chien-Shiung Wu Memorial Hall（吴健雄纪念馆）

The venue of the workshop is in the Lecture Hall of the Chien-Shiung Wu Memorial Hall, in Sipailou Campus of Southeast University. It is at the center of the campus, near the octagon fountain. (See the yellow rectangular mark in the map picture on the following page.)

东南大学四牌楼校区吴健雄纪念馆报告厅（负一楼）

(3) Shing-Tung Yau Center of SEU (丘成桐中心)

The Shing-Tung Yau Center of SEU locates on the 13th & 15th Floor, Yifu Architecture Building in the Sipailou Campus of Southeast University. The building is near the east gate of the campus. (See the orange rectangular mark in the map picture.)

东南大学四牌楼校区逸夫建筑馆 13、15 楼丘成桐中心

2. Accomodation

Nanjing voco Oriental Pearl Hotel (南京东方珍珠 voco 酒店)

Address: No. 389 Zhujiang Road, Nanjing 210018, Jiangsu Province（南京市玄武区珠江路 389 号）

The hotel is about 700 meters (10 mins walk) from the campus and it is quite near the No.2 exit of the metro station Fuqiao (浮桥), Line 3.



(Map of the Sipailou Campus of Southeast University)